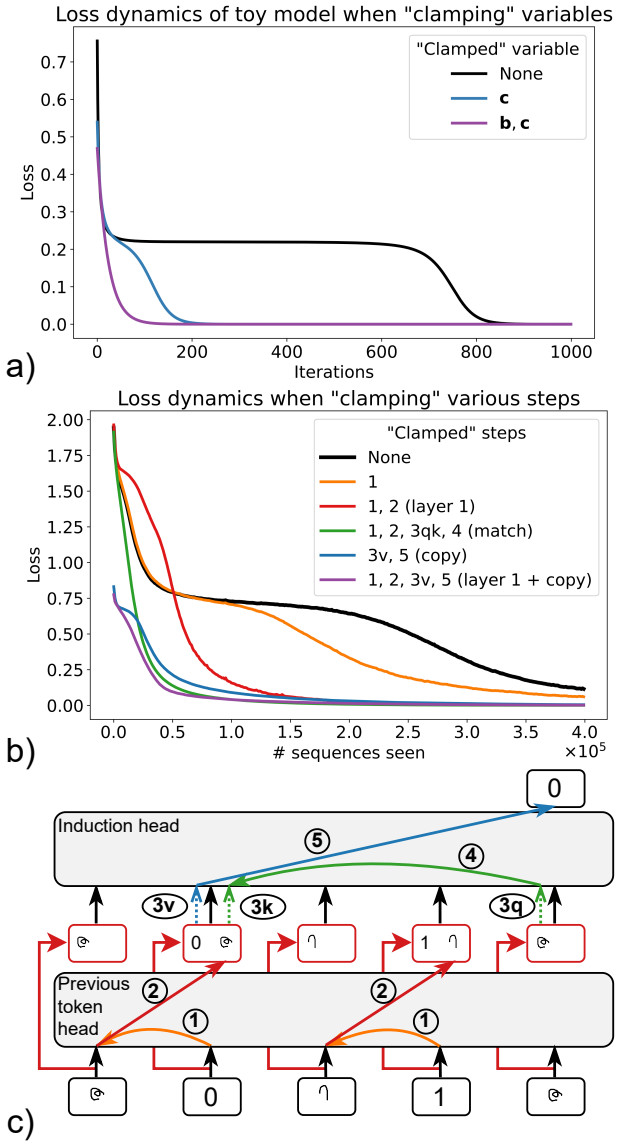

Figure 5. **a)** Loss dynamics when clamping various variables in the toy model presented in Section 4.2. Black shows the learning dynamics when no variable is clamped. Only when all other interacting components ($\mathbf{b}, \mathbf{c}$) are clamped does the loss curve become exponential. **b)** Loss dynamics when clamping various computations outlined in Section 4.1. Black shows the training dynamics of the full network with nothing clamped. **c)** Induction circuit schematic (from Figure 1a), with computation steps labeled. Arrow colors chosen to illustrate which steps are additionally clamped.

to a perfect PT-attend pattern. We see the loss dynamics still exhibit a phase change (orange), with a similar profile, indicating the rest of the computations (2-5) contain interacting subcircuits. Next, we clamp the full Layer 1 computation (PT-attend *and* PT-copy) and ensure that the superimposed residual and output of Layer 1 are disentangled (see Appendix D.1.2 for details). We find that the dynamic profile of the loss shifts substantially (red). While a phase change is still observed, it is much sharper, lasting only $\approx 7.5e4$ iterations (compared to $2e5$, black). This suggests that the formation of Layer 1 attention+output circuits is likely a key sub-component, but there are still interacting sub-components in Steps 3, 4, and 5.

We take this analysis further and clamp the entire match operation (green in Figures 1a, Figure 5), so only the copy operation (blue, Figure 1a) is being learned. We see this loss profile is smoothly exponential, suggesting there are no sub-components hidden within it. We then do the opposite, clamping the copy operation and focusing on the dynamics of match operation formation. We see a small saddle point followed by a quick phase change (lasting $\approx 0.5e5$ iterations), indicating that the match operation should be decomposed further. We do so by clamping both layer 1 and copy components (Steps 1, 2, 3v, 5), which results in a more exponential loss profile (purple, Figure 5).

Given these results, we believe the phase change is primarily determined by three interacting subcircuits:

Subcircuit A: Layer 1 Attending to previous token and copying it forward. Comprised of PT-attend and PT-copy (Steps 1 and 2).

Subcircuit B: IH QK Match Matching queries to keys in the induction head. Comprised of Routing Q, Routing K and IH-match (Steps 3qk and 4).

Subcircuit C: Copy Copying of input label to output. Comprised of Routing V and IH-copy (Steps 3v and 5).

Prior work has often hypothesized the first and second components; our analysis verifies these intuitions and critically identifies the third interacting component through the use of *clamping* (causal manipulations of activations throughout training).

5. Subcircuits can explain data-dependent shifts in phase change timing

To demonstrate the relevance of the identified subcircuits, we turn towards explaining changes in the timing of the phase change as we modify various data properties. Previous work has shown that such data properties can affect the dynamics and emergence of ICL (Chan et al., 2022; Raparthy et al., 2023; Reddy, 2023; Yu et al., 2023). Specifically, we consider two possible variations: increasing the

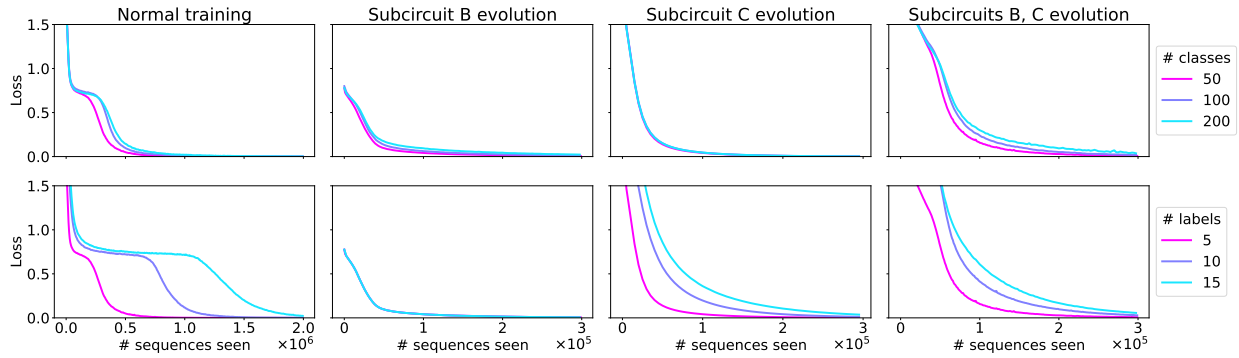

Figure 6. Data-dependent learning dynamics of the induction circuit and assorted subcircuits. Top row shows curves as the # of classes is increased, bottom row shows curves as the # of labels is increased. Left-most column shows training loss over time without any clamps on subcircuits. Middle two columns looks at the evolution of individual subcircuits identified in Section 4.2, by clamping the other two subcircuits. Right-most column shows evolution of a composite circuit (by only clamping Subcircuit A). Middle two plots clearly show how different subcircuits depend on different data properties, and how this can explain the overall difference in learning dynamics.

number of classes C or the number of labels L.

In Figure 6, we find that increasing both the number of classes and number of labels lead to a delay in the formation of induction heads, mostly seen as a shift in the timing of the phase change (leftmost column, Figure 6). This may be because the task becomes more challenging due to the higher data diversity. To better understand what leads to this delay, we look at the effect of these data properties on the formation of some of the subcircuits identified in Section 4, specifically Subcircuits B and C (middle two columns, Figure 6). We find that increasing the number of classes makes learning IH QK Match (Subcircuit B) harder while not increasing the difficulty of Copy (Subcircuit C), thus implying that a delay in Subcircuit B formation causes the delayed phase change. On the other hand, we find that when increasing the number of labels, IH QK Match is learnt as quickly, while copying becomes way harder, indicating that a delay in Subcircuit C formation causes the delayed phase change. These results make intuitive sense—more classes may necessitate learning a higher precision QK matching operation (Subcircuit B), and more labels may necessitate learning a higher precision copying operation (Subcircuit C). Our analysis enables this rigorous decomposition of the phase change of the full model into delays of formation of interacting subcircuits.

To emphasize the importance of considering these subcircuits individually, we consider the joint evolution of both Subcircuits B and C (which we can track by clamping Subcircuit A, analogous to the red line in Figure 5b). This would correspond to just looking at an induction head's formation, when a network is provided with previous token heads (and disentangled outputs of Layer 1). We find that both # classes and # labels seems to affect these dynamics, thus not providing as much clarity on how these different properties might differentially affect learning dynamics. Decomposing into the individual subcircuits that we identified in Section 4.2 provides a better understanding of how data properties differentially increase the difficulty of induction circuit formation, leading to delayed phase changes.

Finally, we note that learning of the individual subcircuits is substantially faster than that of the overall dynamics. By considering data-dependent effects on individual subcircuits, we may be able to predict network behavior without having to train for long amounts of time, waiting for a phase change.

6. Related work

In-context learning From its discovery as an emergent ability in LLMs (Brown et al., 2020), ICL has become a focus of study in NLP (Min et al., 2022; Wei et al., 2023b;a) and in synthetic settings (von Oswald et al., 2023; Chan et al., 2022; Xie et al., 2021). The recent findings of transience from Singh et al. (2023) especially motivated our work, which led to our choice of using a similar setup. Future work could extend our analyses to other setups, like linear regression or natural language tasks.

Induction circuits were first found by Elhage et al. (2021), then further investigated in many settings (Olsson et al., 2022; Wang et al., 2023). Reddy (2023) also considers the dynamics of formation, but uses the correlational approach of *progress measures*. We chose to focus on causal methods (see Appendix D.2) such as clamping, enabling us to discover a different set of 3 subcircuits governing induction circuit learning dynamics (Section 4.2).

Tooling for mechanistic interpretability With the increasing prevalence of LLMs and corresponding safety concerns, mechanistic interpretability has seen a number of toolkits released. Primary open-source frameworks include TransformerLens (Nanda & Bloom, 2022), `nnsight` (Fiotto-Kaufman), and `pyvene` (Wu et al., 2024). All of these

frameworks are built on PyTorch, allow studying and manipulating activations from model checkpionts, and natively support many open-source LLMs. In some cases (Fiotto-Kaufman) they even allow intervention on gradients, but it's unclear if any of these frameworks can be used to manipulate activations *during training with proper propagation of gradients*, which is a crucial and unique feature of our framework. Furthermore, while our library does not support as large a breadth of use cases, we hope the vibrant open-source JAX community will find a mechanistic interpretability framework based in JAX and natively supporting compilation useful (including major speedup benefits), similar to the uptake of the `Rust_circuit` library (Goldowsky-Dill et al., 2023) by Rust users. The only JAX framework the authors are aware of is Tracr (Lindner et al., 2023), which allows researchers to quickly construct transformers that implement certain RASP programs: a complementary focus from TransformerLens, `nnsight`, `pyvene`, `Rust_circuit` and our JAX-based artificial optogenetics framework.

7. Discussion

Our work analyzed induction circuits in two-layer attention-only transformers trained on a synthetic ICL task with a focus on the dynamics of formation. We took inspiration from the original discovery of induction circuits (Olsson et al., 2022) and thus focused on smaller models. While a limitation of our work is the small scale compared to frontier foundation model interpretability (e.g., Wang et al., 2022; Merullo et al., 2024; Palit et al., 2023), we chose this setting due to academic compute constraints and a hope to dive deeper in terms of understanding. Our focus on dynamics, as well as our training setup, was inspired by recent work from Singh et al. (2023) asserting that emergent ICL may be a dynamical phenomenon in settings where multiple competing solutions are possible to solve the task.

Through our analysis, we discovered multiple answers to what does and doesn't need to "go right" for an induction head. We found that induction heads operate additively, with multiple heads used to learn the ICL task more quickly despite not being necessary to solve it. This theme of redundancy carried to previous token heads, which we found to exhibit a many-to-many wiring pattern to induction heads. Next, we used the novel *clamping* methodology to identify three smoothly-evolving subcircuits whose interaction may explain the phase change in induction circuit formation. Furthermore, this better understanding helped us explain data-distribution-dependent changes in phase change timing. Of specific interest here was the quicker timelines of subcircuit formation when isolated, which could serve as useful intuition for understanding the effects various data ablations in large model training may have on the learning dynamics, without having to train full networks on each data ablation from scratch. In terms of specific subcircuits we identified, our work demonstrated the crucial role of the copy subcircuit in explaining data-distributional dependent delays in phase change timing (bottom row, Figure 6). This added understanding may have implications for practitioners when selecting a vocabulary size for LLMs: while larger vocabulary sizes are often preferred due to their increased compression ratio (and thus longer effective context), they may make copying more challenging, thereby delaying induction head formation.

Turning back towards our original motivation from the transience of emergent ICL in transformers (Singh et al., 2023), we hope this improved understanding of key subcircuits within induction circuits, as well as novel analysis tools, can in future shed light on why ICL may fade with overtraining.

Beyond our specific results on induction heads, our work contributes a JAX-based artificial optogenetics framework which supports compilation (yielding nearly a 50x speedup on our hardware, see Appendix B) and can be used to modify activations *during training with proper gradient flow*, a methodology we term clamping. This framework permits causal analyses of learning dynamics, whereas prior work mostly applied causal analyses to end networks or checkpoints from normal training (see Appendix D.2 for a comparison). We view our work as taking the first steps with this approach, and are excited to see how future work may combine our clamping methodology with other causal mechanistic interpretability techniques, such as path patching (Wang et al., 2022; Conmy et al., 2023) or causal mediation analysis (Vig et al., 2020; Geiger et al., 2021; Cao et al., 2021; Geva et al., 2023; Finlayson et al., 2021), throughout training for better understanding of transformer circuit learning dynamics.

Broader impact

This paper presents work whose goal is to advance the field of Machine Learning. There are many potential societal consequences of our work, none which we feel must be specifically highlighted here.

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